

ML Graded Quiz M4

Question 1

Which option below is correct regarding a random forest regressor with n estimators?

- ☐ There are n trees, each developed on the test set.
- ☐ There are n trees, each trained on the training set.
- ☐ There are n trees; therefore, there are $n - 1$ bootstrap samples.
- ☒ There are n trees; therefore, there are n bootstrap samples.

Question 2

Complete the stem:

In standard bagging, which of the following is **true** of base learners?

- ☐ Base learners are constructed on a subset of data instances.
- ☐ Base learners are constructed on all data instances
- ☐ Base learners are constructed on a subset of the feature set and all data instances
- ☒ Base learners are constructed on the entire feature set and all data instances

Question 3

You are given the following training and validation error when developing a Gradient Boosting model. Which `max_depth` should you choose based on the information provided?

max_depth	Training Error	Validation Error
2	121	122
4	118	122
6	121	124
8	118	128
10	112	127

- ☐ 10
- ☐ 4
- ☐ 2
- ☒ 6

Question 4

We apply AdaBoost to a binary classification problem. Suppose we have the below information of actual target classes against predicted classes at a certain iteration. Which option represents the performance or Amount of Say?

Actual	Predicted	Sample weight
1	0	0.1
0	1	0.1

1	1	0.3
0	0	0.2
0	0	0.3

- ☐ 0.7
- ☐ 0.4
- ☐ 0.2
- ☒ 0.6

Question 5

Suppose we have a binary class problem using a random forest classifier with 3 trees using majority voting. Each tree has a 60% accuracy score. What is the minimum accuracy the random forest classifier can have? (Hint: Use 5 observations to determine this)

- ☐ 60%
- ☐ 30%
- ☐ 20%
- ☒ 40%

Question 6

Ensemble learning is based on what?

- ☐ Developing all learners in parallel to form a final model
- ☐ Developing all learners sequentially to form a final model
- ☐ Combining many weak learners to perform better than a single strong learner
- ☒ Combining strong learners to form a single model

Question 7

Suppose you are developing a gradient boosting regressor and have the training data and predictions at the first iteration shown by the columns below. Note that y_{train} are the actuals and $\hat{y}_1$ are the predictions at iteration 1. What vector of values would be the target values to train the base learner at iteration 2 (y_{train2})? Assume a learning rate of 1.

y_{train}	$\hat{y}_1$
502	500
480	495
495	488
482	489

- ☒ y_{train2}
- 500
- 495
- 488
- 489

☐ y_{train2}
1,00
0,97
1,01
0,99

☐ y_{train2}
502
480
495
482

☒ y_{train2}
2
-15
7
-7

Question 8

Which statement is **true** regarding an AdaBoost classifier?

- ☐ Correct classifications are given higher weighting in the next iteration.
- ☐ Incorrect classifications are given lower weighting in the next iteration.
- ☐ Incorrect classifications are given higher weighting in the next iteration.
- ☒ All weights are adjusted by the same factor to ensure an equal chance of being selected at any iteration.

Question 9

Consider the below code in Python. What meta model is used for the stacking model?

```
clf1 = DecisionTreeClassifier()

clf2 = SVC(kernel= 'rbf')

clf3 = GaussianNB()

est_rs = [('DTree', clf1),

('SVM', clf2),

('NB', clf3)

]

mylr = LogisticRegression()
```

```
stackingCLF = StackingClassifier(estimators=est_rs, final_estimator=est_lr,
stack_method='auto', cv=3)
```

- ☐ Decision tree classifier
- ☐ Support vector classifier
- ☐ Gaussian Naïve Bayes
- ☒ Logistic regression

Question 10

In stacking, the splitting of the training data into k -folds is done using what?

- ☐ Random sampling with replacement
- ☐ Systematic sampling
- ☐ Stratified sampling
- ☒ Bootstrap sampling

Question 11

Choose the correct option that answers the below question:

What is the loss function below known as?

$$L = \sum_i^N |y_i - \hat{y}_i|$$

for y and $\hat{y}$ the actual and predicted values respectively over all observations i .

- ☐ L0 loss function
- ☐ L2 loss function
- ☐ L1 loss function
- ☒ L3 loss function

Question 12

Which option is **true** about AdaBoost and Random Forest?

- ☐ In AdaBoost, base learners are dependent on each other whereas in a Random forest model the base learners are independent.
- ☐ In AdaBoost, base learners are independent of each other whereas in a Random forest model the base learners are dependent.
- ☐ AdaBoost and Random forest require tree-based learners.
- ☒ AdaBoost reduces complexity whereas in a Random Forest model base learners improve upon the previous learner, increasing complexity.

Question 13

Suppose we have a binary class problem using a random forest classifier with 3 trees using majority voting. Each tree has an 80% accuracy score. What is the lowest accuracy the random forest classifier can have? (Hint: Use 10 observations to determine this)

- ☐ 80%
- ☐ 70%
- ☐ 20%
- ☒ 50%

Question 14

Which statement is **false** regarding bagging?

- ☐ Bagging models are weighed based on their performance.
- ☐ Bagging relies on resampling.
- ☐ Bagging is not restricted to decision trees but often is due to fast training times and well-established default parameter settings.
- ☒ Bagging can be applied to neural networks.

Question 15

In Gradient boosting, we have predictions $F_j(\mathbf{x})$ at iteration j given features $\mathbf{x}$ as input. An added estimator is represented by h .

$$F_T(\mathbf{x}) = F_0 + \sum_j^{T-1} h_j(\mathbf{x}),$$

What is the learning rate in the above equation?

- ☐ 0
- ☐ 0.5
- ☐ 1
- ☒ Learning rate cannot be obtained from the equation.

Question 16

We apply AdaBoost to a binary classification problem. Suppose we have the below information of actual target classes against predicted classes at a certain iteration. Which option shows the correct adjusted weights before normalizing?

Observation	Actual	Predicted	Sample weight
1	1	0	0.1
2	0	1	0.1
3	1	1	0.3
4	0	0	0.2
5	0	0	0.3

☒

observation	adj weight
1	0.283
2	0.283
3	0.464
4	0.310
5	0.464

☐

observation	adj weight
1	0.135
2	0.135
3	0.405
4	0.270
5	0.405

☐

observation	adj weight
1	0.074
2	0.074
3	0.222
4	0.148
5	0.222

☒

observation	adj weight
1	0.2
2	0.2
3	0.15
4	0.1
5	0.15

SUBMIT